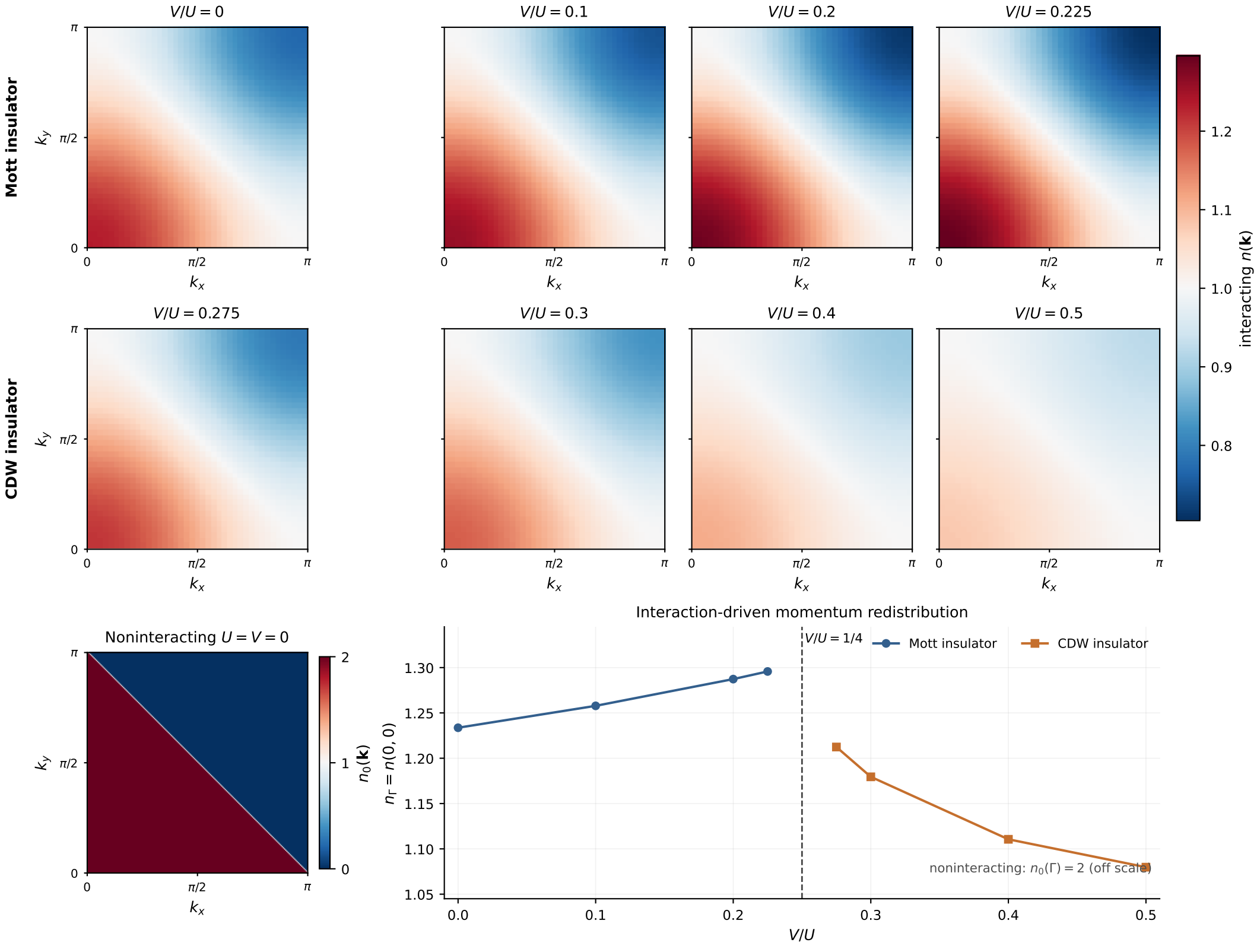
Mott insulator
CDW insulator
$V/U = 0$
$V/U = 0.1$
$V/U = 0.2$
$V/U = 0.225$
$V/U = 0.275$
$V/U = 0.3$
$V/U = 0.4$
$V/U = 0.5$
k_x
k_y
0
π/2
π
interacting $n(\mathbf{k})$
0.8
0.9
1.0
1.1
1.2
Noninteracting $U = V = 0$
$n_0(\mathbf{k})$
0
1
2
Interaction-driven momentum redistribution
$n_\Gamma = n(0,0)$
V/U
$V/U = 1/4$
Mott insulator
CDW insulator
noninteracting: $n_0(\Gamma) = 2$ (off scale)
1.05
1.10
1.15
1.20
1.25
1.30
0.0
0.1
0.2
0.3
0.4
0.5